



PCOD Companion

Project Summary & Design Documentation

A calm, intelligent Android wellness companion for women managing PCOD

The app **observes patterns, offers gentle suggestions, and provides emotional support** — designed to feel like a friend, not a clinical tracker.

1 What the App Does

Daily wellness logging

Track everything that matters in one place — water, sleep, mood, food, exercise, stress, symptoms, medications, emotional state. Each field uses preset chips (≤ 2 taps to log) so the experience never feels like a form.

Personalized weekly plan

Build a routine of recurring tasks across **Diet, Exercise, and Lifestyle** categories. Tasks can be assigned to specific weekdays. A built-in generator creates a starter PCOD-friendly routine based on wake/sleep time, diet preference, and goal.

Cycle tracking

Log period start/end dates, flow level, and symptoms. Empty-state UI gently invites first entry.

Pattern detection (Insights)

Six observational patterns auto-detected from logged data — never prescriptive, always gentle:

- Sleep ↔ Mood
- Sugar ↔ Acne
- Exercise ↔ Energy
- Hydration ↔ Energy
- Stress ↔ Sleep quality
- Protein ↔ Mood

History & Summary

Browse past days as a calendar. Weekly summary with stats plus a **doctor-export feature** for sharing wellness reports.

2 Screens & What They Do

Screen	Purpose
Today	Main hub — focus card, cycle recommendation, insight, plan suggestion, end-of-day closure, emotional check-in, today's progress, plan checklist, daily movement, mini wellness break, quick track tiles, symptoms, medications, weekly reflection
Plan Builder	Build/edit weekly routine with category-tinted cards, day-pill tabs, history hint, generator, snackbar feedback
Cycle Tracker	Log period dates and symptoms
Summary	Weekly stats + doctor export + gear icon → Settings
History	Calendar view of past logs
Settings	Profile, theme, Quiet Mode, reminders, backup/restore, send feedback

3 Tech Stack

- **Language & build:** Kotlin 2.2.10 · AGP 9.2.0 · Gradle 9.4.1 · JDK 21 · minSdk 26 / targetSdk 34
- **UI:** Jetpack Compose + Material 3 (compose-bom 2024.04.00) · Navigation Compose 2.7.7
- **DI:** Hilt 2.59.2 + Hilt-Work 1.2.0
- **Persistence:** Room 2.8.4 (schema **v13**, `exportSchema = false`)
- **Background work:** WorkManager 2.9.0
- **Architecture:** Single activity · Compose-only · MVVM with repository pattern

4 Engagement Features

Reward presence, not perfection

- **End-of-day closure card** — first open of each day, gently summarizes yesterday based on logged data, non-judgmental at every completion level
- **Streak tracking** without shaming on broken streaks

- **Soft failure feedback** — empty states use floral language ("A fresh start ✨") never "no data" or "error"

Make every interaction feel light

- **Optimistic UI updates** — checkboxes flip instantly, DB write happens in background
- **Haptic feedback** on every state-mutating tap
- **Micro-interactions** — press scale, accent border highlight, smooth list reorders, animated day-pill tabs
- **Snackbars not popups** — rotating gentle messages ("Your week looks balanced 🌱")

Adaptive reminders

Three tiers (Low/Medium/High) for morning, hydration, evening check-ins, plus a daily re-engagement worker. Workers are smart — they skip if the user already engaged today, back off when ignored repeatedly (3+ ignores → every other day, 5+ → Mondays only), and reset when the user opens the app again.

Quiet Mode

A dedicated toggle that hides all suggestion cards (Focus, Insight, Plan Suggestion, Closure) and silences feedback snackbars. Logging and tracking remain unaffected. Reminder workers respect Quiet Mode — except the re-engagement nudge (intentional safety net).

First-launch guidance

- Notification permission prompt with reasoning
- First-insight helper card explaining what insights are
- Empty-state CTAs throughout

Local backup & restore

JSON export/import via Storage Access Framework. Versioned format (currently `BACKUP_VERSION = 3`) for forward compatibility. Replace mode wipes existing data and inserts from backup; insights regenerate.

5 Emotional Design Philosophy

Principle	How it's applied
Calm over loud	Pastel palette only — pinks, lavenders, peaches, greens. No neons, no alarming reds
Observational, not prescriptive	Insights say "you felt better when..." not "you must do..."
Opt-in everything	Suggestions render as cards with Accept/Dismiss; nothing auto-mutates user data
Reward presence	Closure cards celebrate showing up at all, even at 0%
Soft failure feedback	Floral language replaces clinical words (no "error", "failed", "invalid")
Quiet Mode aware	All suggestion cards check <code>state.quietMode</code> before rendering

6 Design System

Codified in `.claude/skills/pcod-design/SKILL.md`

Color palette

PastelPink — primary

PastelLavender — secondary

PastelPeach — tertiary

PastelGreen — Diet

Full dark-mode mirror palette also defined.

Category visual identity

Category	Emoji	Background	Accent
Diet	🥗	PastelGreenLight	PastelGreenDark
Exercise	🧘	PastelLavenderLight	PastelLavenderDark
Lifestyle	🌿	PastelPeachLight	PastelPeachDark

Motion specs

- Press scale: 140ms tween, 1f → 0.97f
- Highlight border: 200ms tween, accent at 55% alpha
- List reorder/insert: 350ms tween via `animateItemPlacement()`

- Color/progress tweens: 500–600ms
- Spring scale on selection: `spring(MediumBouncy, MediumLow)`

Spacing & shape

Cards 16–22dp corners, chips 6–14dp · Vertical rhythm 4/8/12/16/24dp · 44dp minimum touch target ·

Pills via `RoundedCornerShape(percent = 50)`

Critical rule (we've shipped this bug twice — third time = unforgivable):

Cards with **fixed light pastel backgrounds** (Today's Focus, all category cards) must use the constant `OnCategoryPastel ( Color(0xFF2D2235) )` for text — **never**

`MaterialTheme.colorScheme.onSurface`, which flips to near-white in dark mode and becomes invisible.

7 Database Discipline

Room **strictly validates** entity vs. DB schema after migration. Mismatched defaults crash the app on startup.

When adding a column:

1. Add field to `@Entity` with `@ColumnInfo(defaultValue = "...")` matching migration SQL **exactly**
2. Bump `version` in `@Database(...)`
3. Add `MIGRATION_OLD_NEW` object in `AppDatabase.companion`
4. Register in `AppModule.provideDatabase()` `.addMigrations(...)`
5. Update `BackupManager` toJson/fromJson + bump `BACKUP_VERSION` if format changes

Conventions: Booleans → INTEGER `"0"` · Strings → TEXT `" "` · Numbers → INTEGER/REAL · Sentinel "not measured" → INTEGER `"-1"`

Current schema (v13)

- `daily_logs` — date, mood, symptoms, water, sleep, exercise, fruit, sugar, stress, sleepQuality, protein, medications, streak, emotionalCheckIn, **planCompletionPct** (newest)
- `cycle_entries` — startDate, endDate, flowLevel, symptoms, notes
- `plan_items` — title, description, category, isCompleted, orderIndex, daysOfWeek
- `insights` — key (unique), message, createdAt, lastShownAt, showCount

8 Architecture Conventions

- **One activity, Compose-only.** No fragments.

- **MVVM** — each screen has `XxxScreen` + `XxxViewModel` (`@HiltViewModel`). State via `MutableStateFlow<XxxUiState>` .
- **Repository pattern** — ViewModels never touch DAOs directly; go through `PCODRepository` / `SettingsRepository` .
- **One-shot UI events** via `MutableSharedFlow` collected with `LaunchedEffect` .
- **Optimistic updates** in `TodayViewModel.updateLogOptimistic { ... }` .
- **Quiet Mode chokepoint** — `TodayViewModel.emitFeedback()` is the only way to fire feedback snackbars.
- **Reminder schedule chokepoint** — `WorkManagerHelper.applyReminderSchedule()` is the single entry point for adaptive reminder workers.

9 What's Already Built

Foundation ✓

- Splash screen + Compose host
- Bottom navigation with 5 tabs
- Light + Dark + System theme support
- Pastel design system + Material 3 theme

Data layer ✓

- Room v13 with 11 migrations
- Backup/restore (JSON, versioned)
- Settings repository (SharedPreferences)
- File provider for sharing exports

Today screen ✓

- Daily focus card, cycle recommendation, insight rotation
- Emotional check-in, progress visualization, plan checklist
- Quick track tiles (water, sleep, fruit, etc.)
- Symptoms, medications, weekly reflection
- End-of-day closure (gated by `lastClosureShownDate`)
- First-insight helper, first-launch reminder prompt

Plan Builder ✓ (*most polished, recent focus*)

- Category-tinted cards with emoji avatars, day chips, edit + drag handles
- Animated pill day-tabs with per-day emojis (🌱 🌸 🌺 🌻 ✨ 🌸 🌙)
- Press scale + highlight micro-interactions

- Smooth `animateItemPlacement` on reorders
- Empty state with CTA button
- Rotating snackbar feedback after add/edit/generate
- Day-history hint ("You usually complete X% of your Tuesday plan")
- Routine generator (wake/sleep time + diet pref + goal)

Other screens ✓

- **Cycle Tracker** — Add/edit cycle entries with friendly empty state
- **Summary** — Weekly stats, doctor export, gear icon → Settings
- **History** — Calendar view with filled-circle indicators
- **Settings** — Profile, theme, Quiet Mode, reminders, backup/restore, feedback

Background work ✓

- **DailyResetWorker** (midnight) — creates today's empty log, snapshots yesterday's plan completion, resets plan checkboxes
- Morning/Hydration/Evening reminder workers (smart guards skip when redundant)
- **Re-engagement worker** (11 AM, fires only after 2-day silence, exempt from Quiet Mode)

Persistent context ✓

- `CLAUDE.md` — project conventions and gotchas
- `.claude/skills/pcod-design/SKILL.md` — full design system spec
- Auto-memory entries for user workflow + Room migration discipline

10 Things Worth Knowing (Gotchas)

Topic	What to remember
AGP 9.x with <code>compileSdk = 34</code>	Unusual combo — works but expect warnings. Don't bump <code>compileSdk</code> without checking other deps
WorkManager init	App implements <code>Configuration.Provider</code> and disables the default initializer in manifest. Don't re-enable it
<code>scheduleDailyReset()</code>	Runs in <code>Application.onCreate()</code> — keep fast, no blocking work
Splash theme	<code>Theme.PCODCompanion.Splash</code> extends <code>Theme.SplashScreen</code> . Don't change activity theme without updating both styles
Room destructive fallback	Rebuilds the table — fine in dev, but does NOT help with column-default validation mismatches . Always match defaults exactly
Backup format	Versioned. Update toJson/fromJson AND bump <code>BACKUP_VERSION</code> if format changes incompatibly
FileProvider authority	<code>\${applicationId}.fileprovider</code> , paths in <code>res/xml/file_paths.xml</code>
Adaptive reminders single entry	Always go through <code>WorkManagerHelper.applyReminderSchedule()</code> . Workers themselves decide whether to fire
Re-engagement is exempt from Quiet Mode	Intentional — it's the safety net for inactive users
Insights are derived data	Wiped + regenerated on backup-import-replace and Reset All Data
Quiet Mode chokepoint	In <code>TodayViewModel</code> , ALWAYS use <code>emitFeedback(msg)</code> — never <code>_feedbackEvents.tryEmit</code> directly
Plan completion tracking is forward-looking	Pre-existing logs have <code>planCompletionPct = -1</code> . Real percentages accumulate going forward via the midnight worker. Day-history hint hides until ≥ 3 measured logs exist
Notification icon must be monochrome	Status bar tints small icons to a single color. Use <code>ic_notification.xml</code> (white-on-transparent leaf vector)

11 What Could Still Be Improved

High-impact polish

- Add/Edit/Generator bottom sheets still use plain category chips — should use category emoji + colors
- Day picker inside Add/Edit sheets uses small grey FlowRow chips with no haptic — should reuse the new `DayPill` composable
- Generator dialog uses `OutlinedTextField` for time entry — would feel warmer with chip presets ("6 AM", "7 AM", "8 AM")
- Drag-to-reorder on plan items — drag handle icon is visual only; needs gesture plumbing or the `reorderable` library

Branding

- Launcher icon still uses the JPEG hack — long-term should be a proper adaptive icon with vector-traced "P+leaf" logo as the foreground
- The placeholder star in `ic_launcher_foreground.xml` is unused; can be deleted once a real foreground exists

Possible new features

- Lock screen widget for quick-track water/mood
- Weekly summary push notification (opt-in, respects Quiet Mode)
- Onboarding carousel for first-time users
- Cycle prediction algorithm based on logged history

12 How to Build & Run

1. Open in Android Studio, plug in phone, hit **Run**
2. For schema/db crashes during dev: uninstall first via `adb uninstall com.pcodcompanion`
3. Logcat filter for crashes: `package:mine level:ERROR`

TL;DR

A **production-ready Kotlin/Compose Android app** for PCOD wellness companionship with **6 main screens, 13-version Room database, adaptive reminder system, JSON backup/restore, dark mode, Quiet Mode, persistent insights, end-of-day closure, and a fully codified design system**. The defining quality is its **emotional design** — every interaction is calm, opt-in, and non-judgmental. Most recent work focused on the Plan Builder, which now has the most polished UI in the app and serves as the reference implementation of the design system.

Generated with care for PCOD Companion 🌸
This document reflects the project state as of the most recent build.